

QAIROON HAIRITI, Oman

WMO: 413150

Lat: 17.253N

Lon: 54.089E

Elev: 881

StdP: 91.18

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	11.8	-10.6	1.7	16.4	-7.8	2.2	18.8	22.9	14.5	18.9	14.7	9.4	340	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.3	33.1	16.8	31.2	17.9	29.9	18.4	24.5	27.9	23.8	26.8	23.3	25.9	5.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.5	26.1	23.0	19.8	25.3	22.6	19.3	24.5	79.7	28.1	76.8	27.1	74.6	26.1	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.0	12.9	11.3	DB	8.2	38.1	0.9	1.2	7.5	39.0	7.0	39.7	6.4	40.4	5.8	41.2	
			WB	1.8	25.1	1.2	0.7	1.0	25.6	0.3	26.0	-0.3	26.4	-1.2	26.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	17.0	18.3	21.4	23.9	25.1	25.2	21.6	20.8	22.2	23.4	21.4	18.1
	DBStd	3.24	2.07	2.19	2.06	2.06	1.91	2.52	1.73	1.54	1.98	1.80	1.51	2.08
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	102	48	23	1	0	0	0	0	0	0	0	1	29
	CDD10.0	4211	217	232	352	416	469	457	361	334	365	416	341	251
	CDD18.3	1271	7	22	95	166	211	207	102	76	115	158	91	22
	CDH23.3	6099	7	46	361	1012	1449	1432	156	79	447	849	238	24
	CDH26.7	1632	0	2	48	249	439	552	53	18	98	156	14	0
Wind	WSAvg	6.3	6.1	6.3	6.2	6.5	6.3	6.5	8.0	7.5	5.6	4.7	5.1	6.5
Precipitation	PrecAvg	114	1	1	3	6	9	8	38	33	7	2	5	1
	PrecMax	287	6	7	30	67	156	36	108	89	76	23	76	25
	PrecMin	23	0	1	0	0	0	0	6	4	0	0	0	0
	PrecStd	68	1	1	8	13	31	10	28	22	15	5	16	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.3	26.4	29.9	32.2	35.4	37.2	32.0	29.5	31.8	30.8	28.2	25.2
		MCWB	14.0	14.4	15.3	15.6	16.3	16.7	25.2	22.4	17.6	16.1	15.4	15.4
	2%	DB	22.6	24.7	28.0	30.7	32.3	34.2	26.9	25.1	29.2	29.5	26.8	23.9
		MCWB	14.3	14.0	14.6	15.8	18.0	16.2	23.2	21.9	17.4	17.2	15.7	15.0
	5%	DB	21.5	23.4	26.6	29.4	30.4	31.8	24.3	23.2	27.4	28.4	25.7	22.9
		MCWB	14.5	14.2	14.4	16.4	19.3	18.1	22.5	22.0	19.0	17.5	15.9	14.8
	10%	DB	20.5	22.1	25.4	28.1	28.8	29.7	22.9	22.1	25.7	27.3	24.7	21.8
		MCWB	14.2	14.2	14.3	17.1	20.1	20.4	22.0	21.5	20.0	17.4	16.0	14.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	19.6	20.7	22.6	24.5	25.2	24.4	23.9	24.3	26.9	23.6	20.9
		MCDB	20.5	22.0	24.1	26.2	28.8	29.6	27.7	27.2	26.8	28.5	24.9	22.1
	2%	WB	17.1	18.3	19.9	21.9	23.6	24.5	23.4	22.8	23.1	23.4	21.0	18.9
		MCDB	19.6	21.0	22.4	25.2	27.4	28.2	25.3	24.2	25.7	25.5	22.9	20.5
	5%	WB	16.5	17.4	19.3	21.4	23.1	24.0	22.6	22.1	22.4	21.6	19.7	17.7
		MCDB	18.9	20.0	21.7	24.5	26.5	27.1	23.6	22.8	24.4	24.7	21.9	19.9
	10%	WB	16.0	16.7	18.7	20.9	22.7	23.5	22.2	21.6	21.7	20.7	18.8	16.8
		MCDB	18.2	19.2	21.0	23.8	25.8	26.2	22.7	22.0	23.4	23.5	21.1	19.3
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.3	7.4	7.5	7.4	6.3	2.2	2.3	5.5	7.7	7.2	7.2
		MCDBR	7.9	8.3	8.5	8.8	10.0	11.1	5.9	5.2	8.9	8.6	7.9	7.9
		MCWBR	4.2	4.8	5.5	6.1	5.9	7.1	2.6	3.3	6.9	6.1	4.9	4.2
	5% WB	MCDBR	6.6	6.7	6.7	7.1	7.4	6.9	3.9	4.1	6.7	8.0	7.1	7.0
		MCWBR	3.4	3.7	3.9	3.7	3.2	3.8	2.2	2.8	5.2	6.1	5.1	4.6
Clear-Sky Solar Irradiance	taub		0.364	0.383	0.428	0.457	0.534	0.696	0.744	0.667	0.529	0.397	0.379	0.365
	taud		2.256	2.184	2.047	1.995	1.809	1.508	1.465	1.600	1.843	2.191	2.236	2.253
	Ebn at Noon		918	917	884	856	782	658	629	684	784	891	894	904
	Edh at Noon		130	146	173	184	218	292	306	269	210	143	131	127
All-Sky Solar Radiation	RadAvg		5.42	6.21	6.90	7.32	7.38	6.73	5.88	6.01	6.46	6.39	5.56	5.17
	RadStd		0.14	0.14	0.24	0.19	0.22	0.30	0.29	0.17	0.20	0.15	0.19	0.16

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.15	N/A	-1.52	N/A	N/A	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page